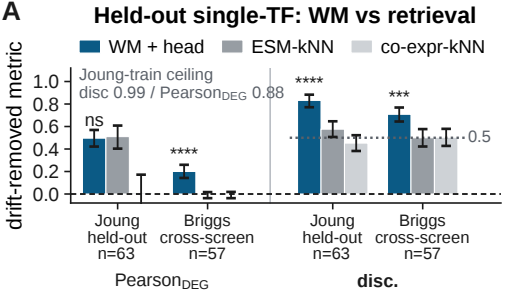
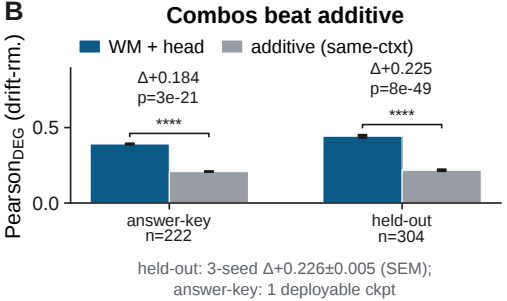
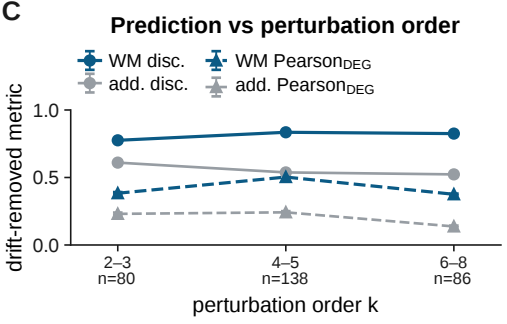
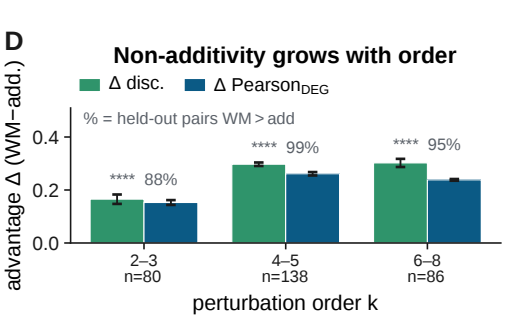

A
Held-out single-TF: WM vs retrieval
WM + head
ESM-kNN
co-expr-kNN
drift-removed metric
Joung-train ceiling
disc 0.99 / Pearson$_{DEG}$ 0.88
ns
****
****
***
0.5
Joung held-out n=63
Briggs cross-screen n=57
Joung held-out n=63
Briggs cross-screen n=57
Pearson$_{DEG}$
disc.


B
Combos beat additive
WM + head
additive (same-ctxt)
Pearson$_{DEG}$ (drift-rm.)
Δ+0.184
p=3e-21
****
Δ+0.225
p=8e-49
****
answer-key n=222
held-out n=304
held-out: 3-seed Δ+0.226±0.005 (SEM);
answer-key: 1 deployable ckpt


C
Prediction vs perturbation order
WM disc.
WM Pearson$_{DEG}$
add. disc.
add. Pearson$_{DEG}$
drift-removed metric
2–3 n=80
4–5 n=138
6–8 n=86
perturbation order k


D
Non-additivity grows with order
Δ disc.
Δ Pearson$_{DEG}$
% = held-out pairs WM > add
advantage Δ (WM−add.)
**** 88%
**** 99%
**** 95%
2–3 n=80
4–5 n=138
6–8 n=86
perturbation order k